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# API Quota Management System

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*Advanced Database Systems (MongoDB)*

## Overview

### Tasks

- Design collections and insert sample data.
- Implement CRUD operations for core entities.
- Write filter and sort queries for operational reporting.
- Build aggregation pipelines for analytics.

### Group Members

|            |                    |
|------------|--------------------|
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## Table of Contents

|                                                     |          |
|-----------------------------------------------------|----------|
| Overview .....                                      | 1        |
| Tasks .....                                         | 1        |
| Group Members .....                                 | 1        |
| <b>Project Introduction .....</b>                   | <b>3</b> |
| Key Features .....                                  | 3        |
| Working & Purpose .....                             | 3        |
| Technology Stack .....                              | 3        |
| Directory Structure.....                            | 4        |
| Directory and File Functions .....                  | 5        |
| Data Model.....                                     | 5        |
| How to run.....                                     | 5        |
| Query Screenshots .....                             | 6        |
| CRUD for usage_bucket, overage_event, api_key ..... | 6        |
| Quota_plans and Developers .....                    | 7        |
| Api Keys with Quota .....                           | 7        |
| Filtering Api Keys .....                            | 8        |
| Aggregation Queries .....                           | 9        |
| 1. Top 10 Developers by Usage.....                  | 9        |
| 2. Simple Active Plan Summary.....                  | 10       |

# Project Introduction

This project implements an **API quota and rate limiting system** using MongoDB. It includes a complete data model plus a set of scripts that you can run directly in **mongosh** to enforce quotas, track usage, handle overages, and maintain full audit traceability.

## Key Features

- **Strict schema validation** for developers, API keys, and quota plans.
- **Time-bucketed usage counters** for per minute, per-hour, and per-day windows.
- **Overage handling**, including throttling, warnings, and key revocation, recorded as events.
- **Audit trail** for operational and security actions in a dedicated **audit\_logs** collection.
- **Realistic sample dataset** with 25 developers across 4 plan tiers, including active and revoked keys.

**Deliverable:** Ready-to-run mongosh examples covering CRUD, filter/sort reporting, and aggregation analytics.

## Working & Purpose

The purpose of this project is to provide a structured **API Quota Management System** that controls and monitors how clients consume an API, ensuring fair usage, stability, and protection against abuse. It works by storing developers and their subscription **quota plans**, issuing **API keys** that link each consumer to a plan, and recording every request into time-based **usage buckets** (per minute, hour, and day). For each request, the system checks the current bucket counters against the plan limits; when a limit is reached or exceeded, it creates an **overage event** and applies an enforcement action (such as warning, throttling, or revocation). All operational and security-relevant actions are also written to an append-only **audit log**, making the system traceable and easy to report on using filter/sort queries and aggregation pipelines in MongoDB.

## Technology Stack

| Component  | Technology                        |
|------------|-----------------------------------|
| Database   | MongoDB                           |
| Shell      | mongosh                           |
| Language   | JavaScript (supported by mongoDB) |
| Validation | MongoDB JSON Schema               |

## Directory Structure

```
api_rate_limit_quota_db
|-- COMPLETE.js
|-- readme.md
|-- Queries/
| |-- aggregation.js
| |-- filter_and_sort.js
| \-- CRUD/
| |-- api_key.js
| |-- audit_log.js
| |-- developer.js
| |-- overage_event.js
| |-- quota_plans.js
| \-- usage_bucket.js
|         usage_bucket.js
|-- sample_data/
| |-- api_keys.js
| |-- audit_logs.js
| |-- developers.js
| |-- overage_events.js
| |-- quota_plans.js
| \-- usage_bucket.js
\-- schema/
    |-- api_keys.js
    |-- audit_logs.js
    |-- developers.js
    |-- overage_events.js
    |-- quota_plans.js
    |-- usage_buckets.js
    \-- schema.md
        overage_events.js
        quota_plans.js
        schema.md
        usage_buckets.js
```

## Directory and File Functions

| Directory / File | Function                                                                                    |
|------------------|---------------------------------------------------------------------------------------------|
| schema/          | Contains one JavaScript file per collection, each defining a MongoDB JSON Schema validator. |
| schema/schema.md | Documentation of every field constraint and enumeration for all six collections.            |
| sample_data/     | Seed scripts that insert realistic documents into each collection.                          |

| Directory / File           | Function                                                                                                                      |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Queries/CRUD/              | CRUD scripts (create, read, update, delete) for each collection, using valid cross-collection references.                     |
| Queries/filter_and_sort.js | Examples such as recently onboarded developers, active keys by plan, top daily usage buckets, and recent enforcement actions. |
| Queries/aggregation.js     | Aggregation pipelines (e.g., top developers by today's usage and active-key plan distribution).                               |
| COMPLETE.js (root)         | Global script. Loads and executes schema/, sample_data/, and Queries/ files in order.                                         |

## Data Model

| Collection            | Purpose                                                                                                                                     |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>developers</b>     | Represents API consumers — individuals or teams. Stores identity, organisation, and account status.                                         |
| <b>quota_plans</b>    | Named plan tiers (Free, Basic, Pro, Enterprise) with rate-limit rules such as requestsPerMinute, requestsPerHour, and requestsPerDay.       |
| <b>api_keys</b>       | Credentials issued to developers. Linked to a developer and a plan. Tracks active/revoked state and timestamps.                             |
| <b>usage_buckets</b>  | Time-windowed request counters per API key. Supports minute, hour, and day period types to enable rolling-window enforcement.               |
| <b>overage_events</b> | Records each policy violation: which key exceeded which limit, by how much, and what enforcement action was taken.                          |
| <b>audit_logs</b>     | Append-only operational and security log. Captures actor (admin/system/developer), action, target entity, timestamp, and relevant metadata. |

## How to run

```
shell> mongosh COMPLETE.js
```

# Query Screenshots

## CRUD for usage\_bucket, overage\_event, api\_key

### 1. Usage Bucket CRUD operations...

Usage bucket for today: null

### 2. Overage Event CRUD operations...

New overage event created for API key: sha256\$a1f9c27088884804ed40a2b3c4d5e6f7081920a3b4c5d6e7f80910a2b3c4d5e6

Overage events from the last 7 days: [

```
{
  apiKeyId: ObjectId('69fcd88cbce8908bcd7c2944'),
  time: ISODate('2026-05-07T18:23:08.562Z'),
  limitType: 'minute',
  allowed: 10,
  observed: 15,
  actionTaken: 'throttle_429'
}
```

]

### 3. API Key CRUD operations...

New API key created for developer: claire@quantleaf.com

Active API keys for Claire: [

```
{
  _id: ObjectId('69fcd88cbce8908bcd7c2924'),
  developerId: ObjectId('69fcd88cbce8908bcd7c2908'),
  keyHash: 'sha256$a1f9c2e0b8d4e7f1a23b4c5d6e7f8091a2b3c4d5e6f70819a2b3c4d5e6f70819',
  planId: ObjectId('69fcd88cbce8908bcd7c2923'),
  active: true,
  createdAt: ISODate('2024-03-03T10:00:00.000Z'),
  revokedAt: null
},
{
  _id: ObjectId('69fcd88cbce8908bcd7c2925'),
  developerId: ObjectId('69fcd88cbce8908bcd7c2908'),
  keyHash: 'sha256$b2e8d3f1c9e5f8021b34c5d6e7f80192b3c4d5e6f7081920b3c4d5e6f7081920',
  planId: ObjectId('69fcd88cbce8908bcd7c2923'),
  active: true,
  createdAt: ISODate('2024-03-03T10:05:00.000Z'),
  revokedAt: null
},
{
  _id: ObjectId('69fcd88cbce8908bcd7c29ab'),
  developerId: ObjectId('69fcd88cbce8908bcd7c2908'),
  keyHash: 'sha256$newkey99887766554433221100ffeeddccbbaa99887766554433221100ffeedd',
  planId: ObjectId('69fcd88cbce8908bcd7c2921'),
  active: true,
  createdAt: ISODate('2026-05-07T18:23:08.589Z'),
  revokedAt: null
}
```

]

## Quota\_plans and Developers

4. Quota Plans CRUD operations...

Active quota plans: [

```
{
  _id: ObjectId('69fcd88cbce8908bcd7c2920'),
  name: 'Free',
  rules: { requestsPerMinute: 10, requestsPerDay: 500, burstLimit: 0 },
  description: 'Free tier for hobby developers. Hard limit enforced, no overage allowed.',
  active: true
},
{
  _id: ObjectId('69fcd88cbce8908bcd7c2921'),
  name: 'Basic',
  rules: { requestsPerMinute: 60, requestsPerDay: 20000, burstLimit: 10 },
  description: 'Basic paid plan for small projects. Burst allowance of 10 extra requests per minute.',
  active: true
},
{
  _id: ObjectId('69fcd88cbce8908bcd7c2922'),
  name: 'Pro',
  rules: { requestsPerMinute: 300, requestsPerDay: 100000, burstLimit: 50 },
  description: 'Pro plan for production applications. Paid overage permitted after daily limit.',
  active: true
},
{
  _id: ObjectId('69fcd88cbce8908bcd7c2923'),
  name: 'Enterprise',
  rules: {
    requestsPerMinute: 2000,
    requestsPerDay: 5000000,
    burstLimit: 300
  },
  description: 'Enterprise plan with dedicated limits and priority support.',
  active: true
}
]
```

5. Developer CRUD operations...

Developer status updated to active for: claire@quantleaf.com

Developers suspended from Kremlin Apps:

## Api Keys with Quota

Active API keys with daily quota >= 100,000, ordered by strictest per-minute limit: [

```
{
  _id: ObjectId('69fcd88cbce8908bcd7c2935'),
  developerId: ObjectId('69fcd88cbce8908bcd7c291e'),
  planId: ObjectId('69fcd88cbce8908bcd7c2922'),
  createdAt: ISODate('2025-10-26T10:05:00.000Z'),
  planRules: { requestsPerMinute: 300, requestsPerDay: 120000, burstLimit: 50 }
},
{
  _id: ObjectId('69fcd88cbce8908bcd7c2934'),
  developerId: ObjectId('69fcd88cbce8908bcd7c291e'),
  planId: ObjectId('69fcd88cbce8908bcd7c2922'),
  createdAt: ISODate('2025-10-26T10:00:00.000Z'),
  planRules: { requestsPerMinute: 300, requestsPerDay: 120000, burstLimit: 50 }
},
]
```

## Filtering Api Keys

```
10 most recently onboarded ACTIVE developers: [
  {
    _id: ObjectId('69fcd88cbce8908bcd7c291f'),
    name: 'Robert Harrison',
    email: 'robert.harrison@protonmail.com',
    organization: 'Indie',
    createdAt: ISODate('2025-12-14T16:00:00.000Z')
  },
  {
    _id: ObjectId('69fcd88cbce8908bcd7c291e'),
    name: 'Grace Thompson',
    email: 'grace.t@maplebridge.ca',
    organization: 'MapleBridge',
    createdAt: ISODate('2025-10-25T12:00:00.000Z')
  },
  {
    _id: ObjectId('69fcd88cbce8908bcd7c291d'),
    name: 'Yusuf Demin',
    email: 'yusuf.demin@bosporlabs.tr',
    organization: 'Bosphor Labs',
    createdAt: ISODate('2025-09-18T11:30:00.000Z')
  },
  {
    _id: ObjectId('69fcd88cbce8908bcd7c291b'),
    name: 'Sven Larsson',
    email: 'sven@nordlys.se',
    organization: 'Nordlys AB',
    createdAt: ISODate('2025-08-30T07:20:00.000Z')
  },
  {
    _id: ObjectId('69fcd88cbce8908bcd7c291a'),
    name: 'Chiomaa Okafor',
    email: 'chiomaa@bayoutech.ng',
    organization: 'Bayou Tech Lagos',
    createdAt: ISODate('2025-07-22T10:45:00.000Z')
  },
  {
    _id: ObjectId('69fcd88cbce8908bcd7c2919'),
    name: 'Liam Walsh',
    email: 'liam.walsh@celticforge.ie',
    organization: 'CelticForge',
    createdAt: ISODate('2025-06-11T08:00:00.000Z')
  },
  {
    _id: ObjectId('69fcd88cbce8908bcd7c2917'),
    name: 'Marcus Reed',
    email: 'm.reed@stellardata.us',
    organization: 'Stellar Data Corp',
    createdAt: ISODate('2025-04-05T14:30:00.000Z')
  },
  {
    _id: ObjectId('69fcd88cbce8908bcd7c2916'),
    name: 'Eva Novak',
    email: 'eva@novak-systems.cz',
    organization: 'Novak Systems',
    createdAt: ISODate('2025-03-12T09:00:00.000Z')
  },
  {
    _id: ObjectId('69fcd88cbce8908bcd7c2914'),
    name: 'Fatima Zahra',
    email: 'fatima.zahra@atlasmena.ma',
    organization: 'Atlas MENA',
    createdAt: ISODate('2025-02-08T11:10:00.000Z')
  },
  {
    _id: ObjectId('69fcd88cbce8908bcd7c2913'),
    name: 'Kenji Yamamoto',
    email: 'kenji.y@sakuralabs.jp',
    organization: 'Sakura Labs',
    createdAt: ISODate('2025-01-19T07:45:00.000Z')
  }
]
```



## Aggregation Queries

### 1. Top 10 Developers by Usage

```
Top 10 developers by today's usage: [  
  {  
    totalRequestsToday: 11436366,  
    name: 'Marcus Reed',  
    organization: 'Stellar Data Corp',  
    plan: 'Enterprise',  
    activeKeyCount: 2  
  },  
  {  
    totalRequestsToday: 9376359,  
    name: 'Sara Khan',  
    organization: 'Helios Cloud',  
    plan: 'Enterprise',  
    activeKeyCount: 2  
  },  
  {  
    totalRequestsToday: 7666675,  
    name: 'Claire Thompson',  
    organization: 'QuantLeaf Analytics',  
    plan: 'Enterprise',  
    activeKeyCount: 2  
  },  
  {  
    totalRequestsToday: 254477,  
    name: 'Mei Lin Chen',  
    organization: 'Pagoda Tech',  
    plan: 'Pro',  
    activeKeyCount: 2  
  },  
  {  
    totalRequestsToday: 174914,  
    name: 'James Smith',  
    organization: 'NimbusForge',  
    plan: 'Pro',  
    activeKeyCount: 1  
  },  
  {  
    totalRequestsToday: 112422,  
    name: 'Grace Thompson',  
    organization: 'MapleBridge',  
    plan: 'Pro',  
    activeKeyCount: 2  
  },  
  {  
    totalRequestsToday: 101234,  
    name: 'David Lee',  
    organization: 'Aurora Systems',  
    plan: 'Pro',  
    activeKeyCount: 1  
  },  
  {  
    totalRequestsToday: 98765,  
    name: 'Emily White',  
    organization: 'Zenith Networks',  
    plan: 'Pro',  
    activeKeyCount: 1  
  },  
  {  
    totalRequestsToday: 87654,  
    name: 'Michael Brown',  
    organization: 'Nebula Labs',  
    plan: 'Pro',  
    activeKeyCount: 1  
  },  
  {  
    totalRequestsToday: 76543,  
    name: 'Sophia Green',  
    organization: 'Cosmos Dynamics',  
    plan: 'Pro',  
    activeKeyCount: 1  
  },  
  {  
    totalRequestsToday: 65432,  
    name: 'Daniel Black',  
    organization: 'Orion Innovations',  
    plan: 'Pro',  
    activeKeyCount: 1  
  }  
]
```

```
{  
  totalRequestsToday: 174914,  
  name: 'James Smith',  
  organization: 'NimbusForge',  
  plan: 'Pro',  
  activeKeyCount: 1  
},  
{  
  totalRequestsToday: 112422,  
  name: 'Grace Thompson',  
  organization: 'MapleBridge',  
  plan: 'Pro',  
  activeKeyCount: 2  
},  
{  
  totalRequestsToday: 101234,  
  name: 'David Lee',  
  organization: 'Aurora Systems',  
  plan: 'Pro',  
  activeKeyCount: 1  
},  
{  
  totalRequestsToday: 98765,  
  name: 'Emily White',  
  organization: 'Zenith Networks',  
  plan: 'Pro',  
  activeKeyCount: 1  
},  
{  
  totalRequestsToday: 87654,  
  name: 'Michael Brown',  
  organization: 'Nebula Labs',  
  plan: 'Pro',  
  activeKeyCount: 1  
},  
{  
  totalRequestsToday: 76543,  
  name: 'Sophia Green',  
  organization: 'Cosmos Dynamics',  
  plan: 'Pro',  
  activeKeyCount: 1  
},  
{  
  totalRequestsToday: 65432,  
  name: 'Daniel Black',  
  organization: 'Orion Innovations',  
  plan: 'Pro',  
  activeKeyCount: 1  
}
```

## 2. Simple Active Plan Summary

```
Simple active plan summary: [  
  {  
    activeKeyCount: 10,  
    planName: 'Pro',  
    active: true,  
    requestsPerMinute: 300,  
    requestsPerDay: 120000,  
    burstLimit: 50,  
    developerCount: 8  
  },  
  {  
    activeKeyCount: 7,  
    planName: 'Basic',  
    active: true,  
    requestsPerMinute: 60,  
    requestsPerDay: 20000,  
    burstLimit: 10,  
    developerCount: 7  
  },  
  {  
    activeKeyCount: 6,  
    planName: 'Enterprise',  
    active: true,  
    requestsPerMinute: 2000,  
    requestsPerDay: 5000000,  
    burstLimit: 300,  
    developerCount: 3  
  },  
  {  
    activeKeyCount: 3,  
    planName: 'Free',  
    active: true,  
    requestsPerMinute: 10,  
    requestsPerDay: 500,  
    burstLimit: 0,  
    developerCount: 2  
  }  
]
```

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